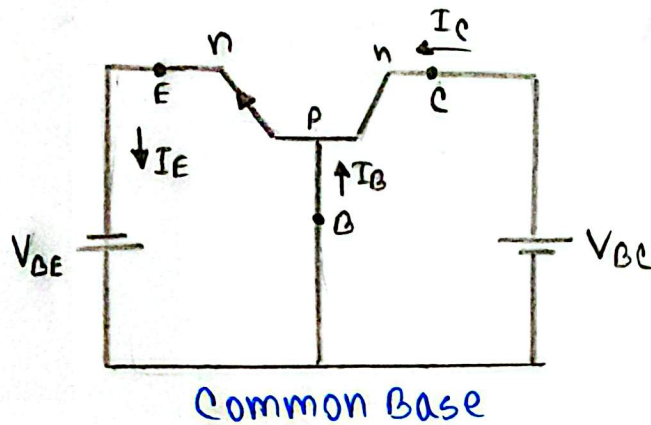
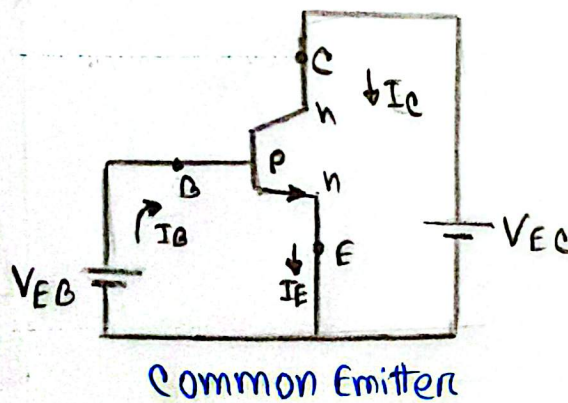


Ans. to the Question No. 01

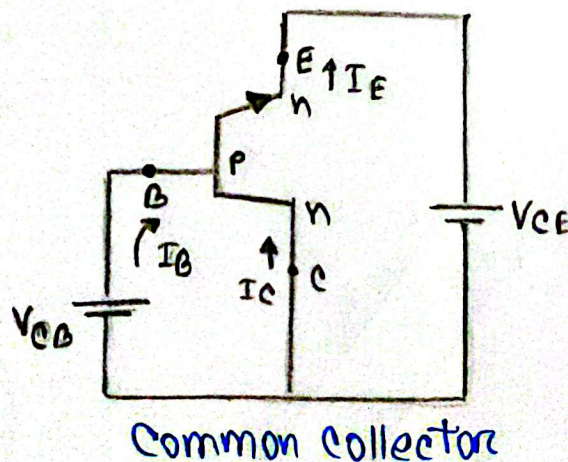
□ Circuit Diagram of Common Base, Common Emitter and common collector —



Here,
BE Junction is forward bias
and BC Junction is reverse bias
Input Current = I_E
Output Current = I_C



Here,
EB Junction is forward bias
and EC Junction is reverse bias
Input Current = I_B
Output Current = I_C

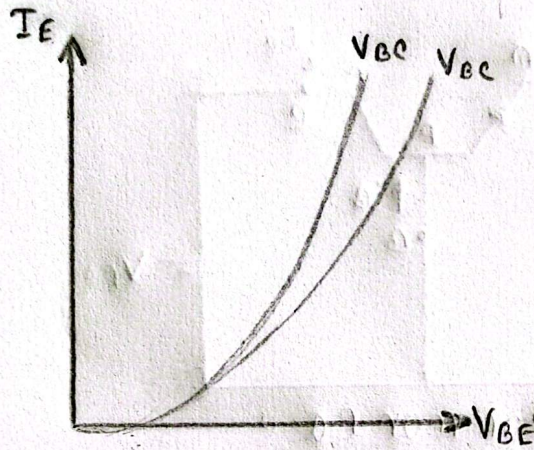


Here,
CB Junction is forward bias
and CE Junction is reverse bias
Input Current = I_B
Output Current = I_E

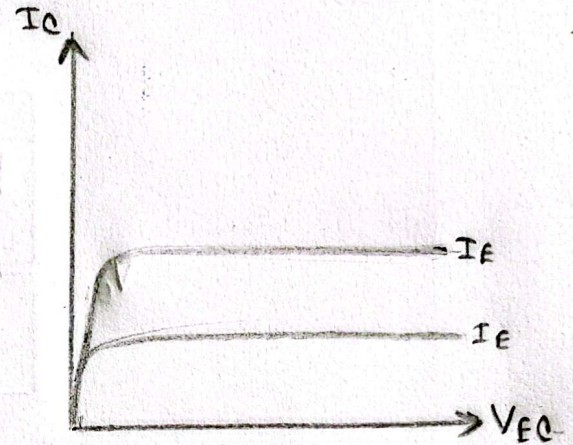
□ I-V characteristics of Common Base, Common Emitter and Common Collector—

Common Base

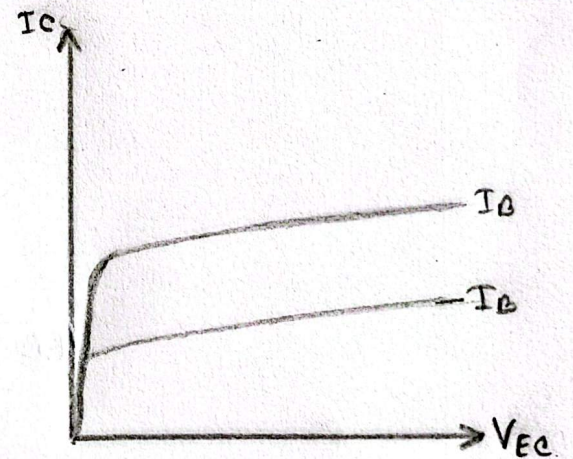
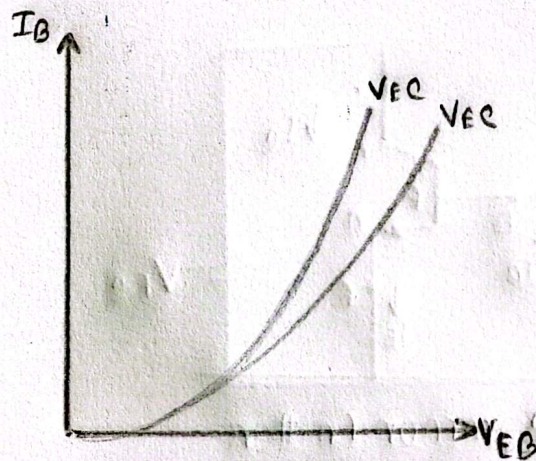
Input



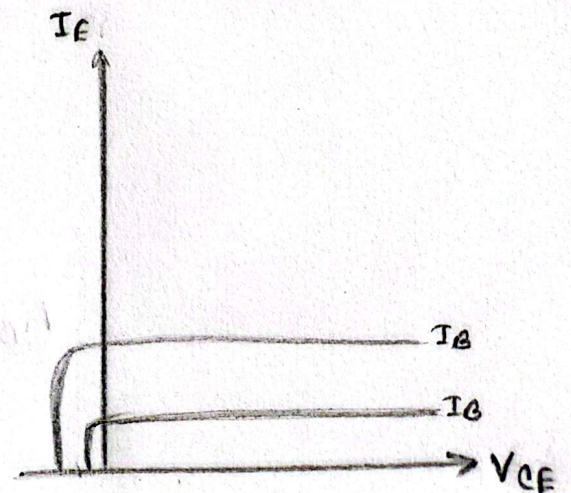
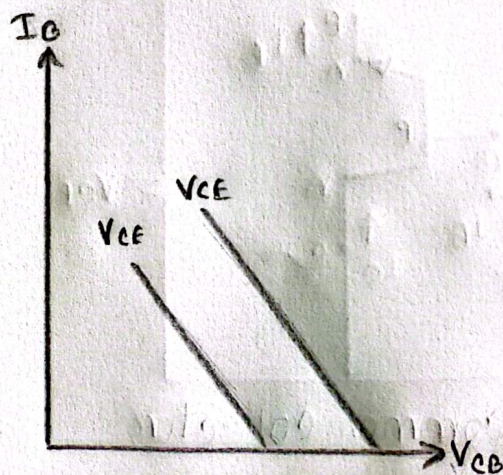
Output



Common Emitter



Common Collector



Ans. to the Question No. 02

We know,

$$I_E = I_B + I_C$$

Now,

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

$$\Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\Rightarrow \alpha = \frac{\beta}{1 + \beta}$$

Again,

$$\frac{I_E}{I_E} = \frac{I_B}{I_E} + \frac{I_C}{I_E}$$

$$\Rightarrow 1 = \frac{1}{\gamma} + \alpha$$

$$\Rightarrow \alpha = \frac{\gamma - 1}{\gamma}$$

Again,

$$\frac{I_E}{I_B} = \frac{I_B}{I_B} + \frac{I_C}{I_B}$$

$$\Rightarrow \gamma = 1 + \beta$$

Thus,

$$\gamma = 1 + \beta = \frac{1}{1 - \alpha}$$

Current gain for;

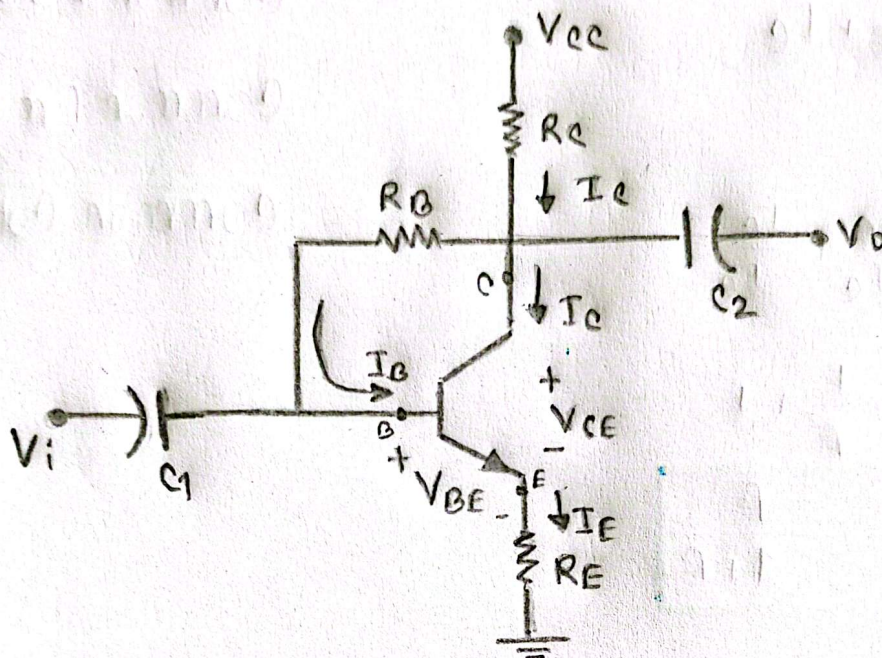
Common Base, $\alpha = \frac{I_C}{I_E}$

Common Emitter, $\beta = \frac{I_C}{I_B}$

Common Collector, $\gamma = \frac{I_E}{I_B}$

Ans. to the Question No. 03

□ Collector-Feedback Configuration-



considering the coupling capacitors as open,
Applying KVL in input side,

$$-V_{CC} + R_C I_C' + R_B I_B + V_{BE} + R_E I_E = 0$$

$$\Rightarrow -V_{CC} + R_C \beta I_B + R_B I_B + V_{BE} + R_E \beta I_B = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$$

$$\begin{aligned} \text{Here,} \\ I_C' &\approx I_C \\ I_E &\approx I_C \\ I_C &= \beta I_B \end{aligned}$$

Applying KVL in output side,

$$-V_{CC} + I_C' R_C + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$$

And,

$$V_{BE} = 0.7 \text{ V}$$

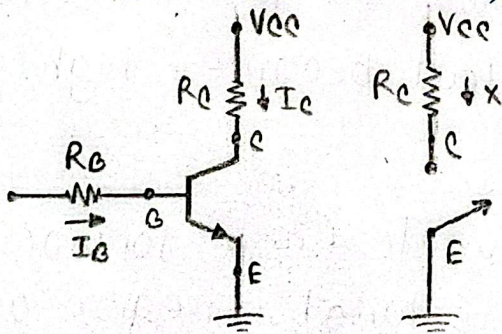
Ans. to the Question No. 04

- A Bipolar Junction Transistor (BJT) is not only used as an amplifier but also widely used as a switch and an inverter. This is possible because a BJT can operate in three distinct regions - Cutoff, Saturation and Active.

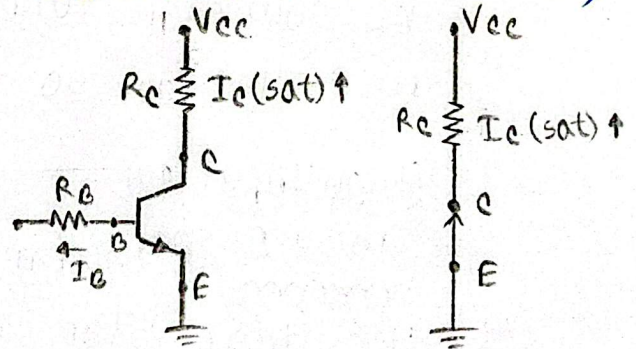
□ BJT as a SWITCH:

When a BJT is connected in Common-Emitter configuration, it behaves like an electronic switch. Basically a BJT switch uses base current (I_B) to control a larger collector-emitter current for loads like LEDs or relays.

• Cutoff (Open Switch)



• Saturation (Closed Switch)



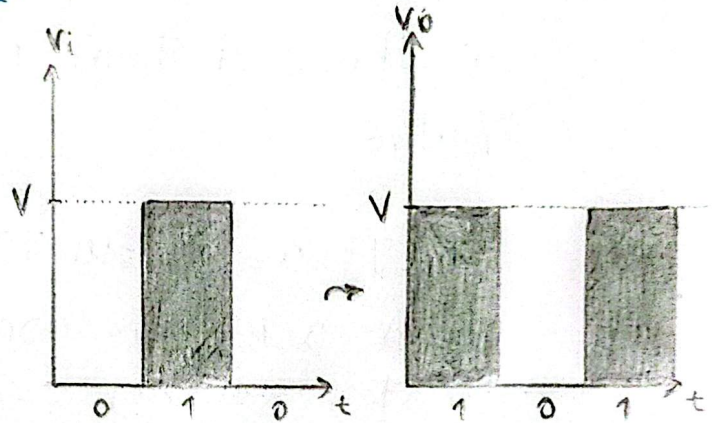
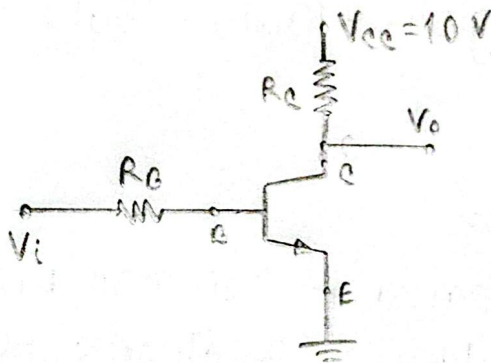
In cutoff region, ($I_B = 0$) I_C is almost zero, blocking load current. As a result, $V_{CE} \approx V_{CC}$ so the transistor works like an open circuit. Thus, switch is OFF.

In saturation region, sufficient I_B drives I_C to maximum ($I_C \approx \beta I_B$), dropping $V_{CE} \approx 0.2V$. Here, Base Resistor limits I_B . $I_B = \frac{V_{in} - V_{BE}}{R_B}$, ensuring saturation doesn't exceed transistor rating. As a result, the transistor works

like a short circuit. Thus, switch is ON.

Input (Base)	Transistor state	Output (collector)
Low (0)	Cutoff	High (1)
High (1)	Saturation	Low (0)

□ BJT as an INVERTER:



(Low Input (0))

Case-1: Base-Emitter junction is not forward bias; thus the transistor is OFF (cutoff region). As a result, no collector current flows and the collector voltage rises to V_{cc} so the output becomes high (1).

(High Input (1))

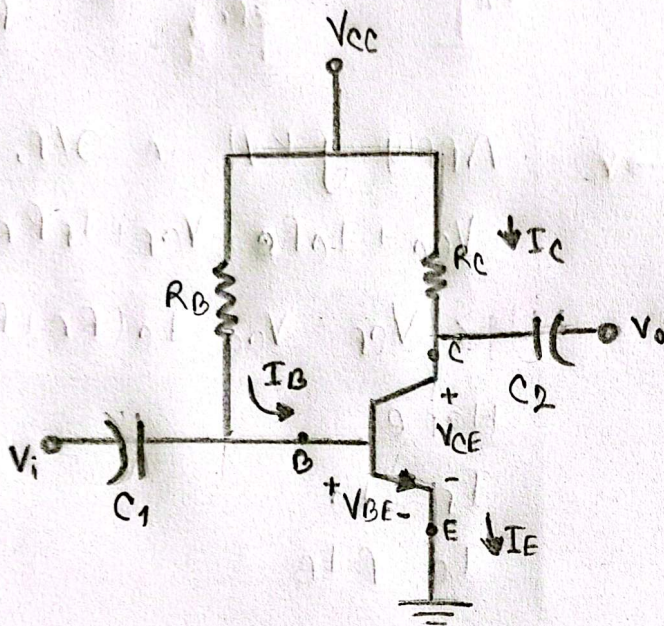
Case-2: Base-Emitter junction is in forward bias; the transistor enters saturation region and a large amount of collector current flows and the voltage becomes nearly 0. Thus, output is low (0).

Input	Output
0	1
1	0

This confirms that, BJT can be used as an inverter

Ans. to the Question No. 05

(a) Fixed Bias



Applying KVL in I/P,

$$-V_{CC} + I_B R_B + V_{BE} = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

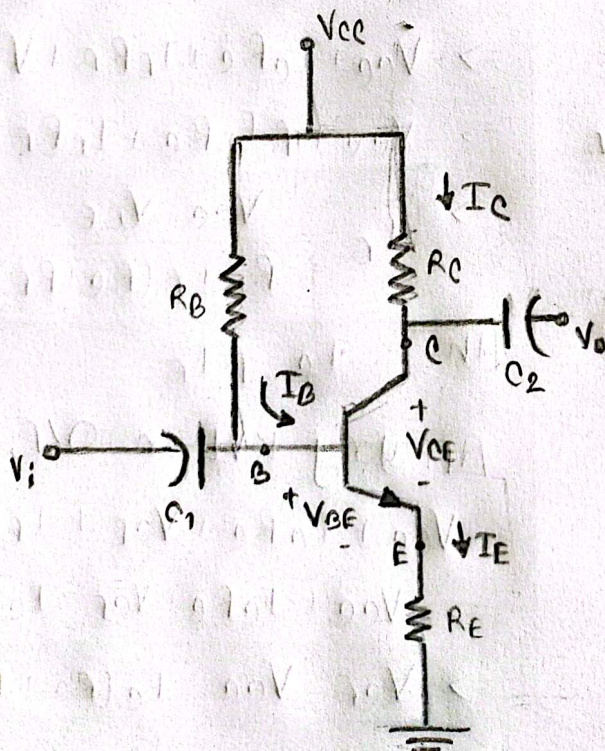
Then, $I_C = \beta I_B$

Applying KVL in O/P,

$$-V_{CC} + I_C R_C + V_{CE} = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C R_C$$

(b) Emitter Stabilized



Applying KVL in I/P,

$$-V_{CC} + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E}$$

Then, $I_C = \beta I_B$

Applying KVL in O/P,

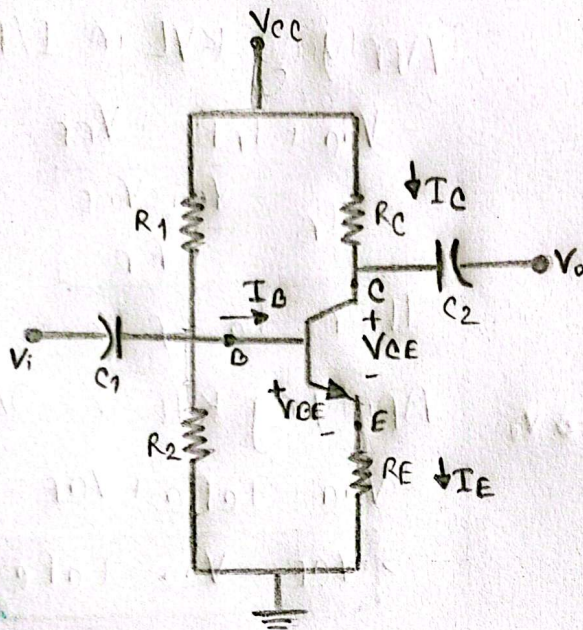
$$-V_{CC} + I_C R_C + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$$

($\because I_E \approx I_C$)

(c) Voltage Divider Bias



Applying KVL in I/P,

$$-V_{TH} + I_B R_{TH} + V_{BE} + I_E R_E = 0$$

$$\Rightarrow I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E}$$

Applying KVL in O/P,

$$-V_{CC} + I_C R_C + V_{CE} + I_E R_E = 0$$

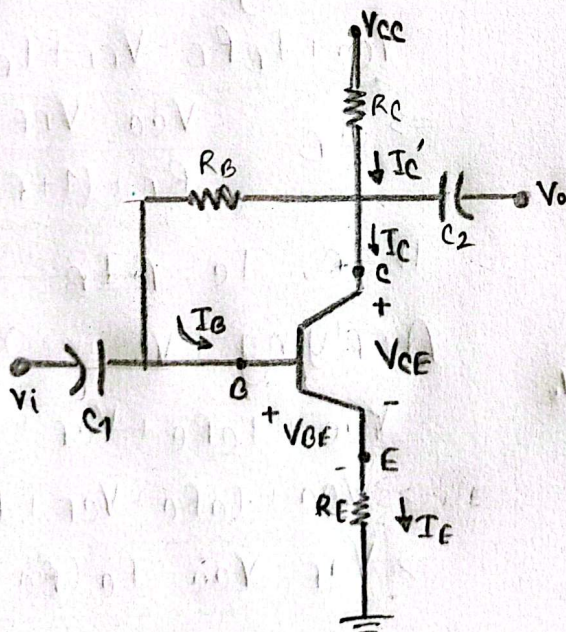
$$\Rightarrow V_{CE} = V_{CC} - I_C (R_E + R_C)$$

Here,

$$I_E \approx I_C$$

$$I_C = \beta I_B$$

(d) Collector Feedback



Applying KVL in I/P,

$$-V_{CC} + I_C' R_C + I_B R_B + V_{BE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + I_B R_B + V_{BE} + I_C R_E = 0$$

$$\Rightarrow -V_{CC} + \beta I_B R_C + I_B R_B + V_{BE} + \beta I_B R_E = 0$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta (R_C + R_E)}$$

Then, $I_C = \beta I_B$

Applying KVL in O/P,

$$-V_{CC} + I_C' R_C + V_{CE} + I_E R_E = 0$$

$$\Rightarrow -V_{CC} + I_C R_C + V_{CE} + I_C R_E = 0$$

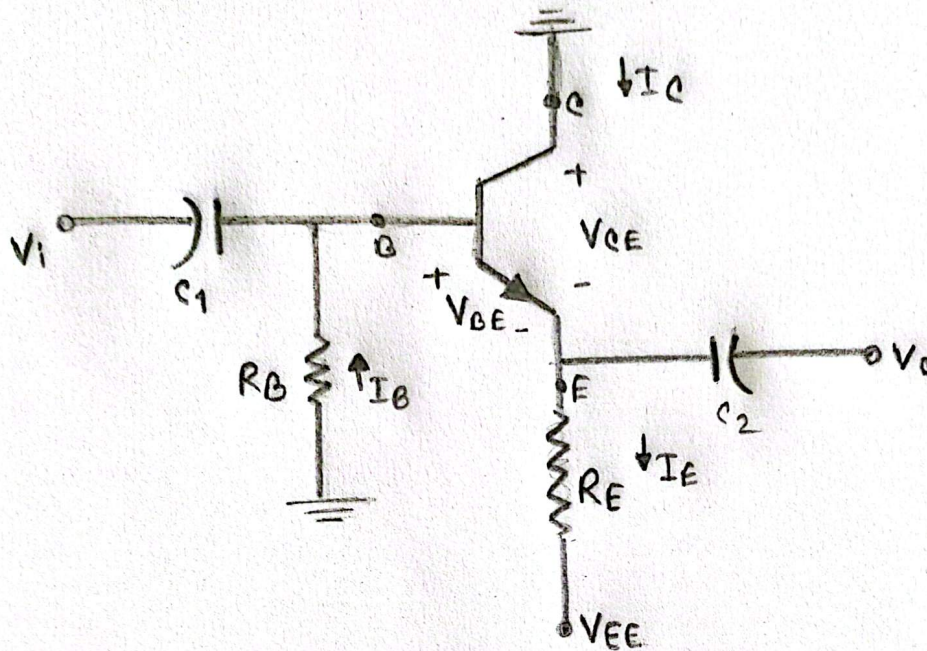
$$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Here,

$$I_C' \approx I_C$$

$$I_E \approx I_C$$

(e) Emitter Follower



Applying KVL in I/P,

$$I_B R_B + V_{BE} + I_E R_E + V_{EE} = 0$$

$$\Rightarrow I_B R_B + V_{BE} + (\beta + 1) I_B R_E + V_{EE} = 0$$

$$\Rightarrow I_B = \frac{-V_{EE} - V_{BE}}{R_B + (\beta + 1) R_E}$$

Then, $I_C = \beta I_B$

Applying KVL in O/P,

$$V_{CE} + I_E R_E + V_{EE} = 0$$

$$\Rightarrow V_{CE} = -V_{EE} - I_E R_E$$

Ans. to the Question No. 06

• In accordance with the given instruction, the values of all circuit elements used in the following problems are taken from the last two digits (79) of my roll.

• Roll: 2403179

Fixed Bias (4.3)

Problem-01

Applying KVL in I/P,

$$-9 + 79 \times 10^3 \times I_{BQ} + V_{BE} = 0$$

$$\Rightarrow I_{BQ} = 1.05 \times 10^{-4} \text{ A} \quad \underline{\underline{\text{Ans.}}}$$

$$\begin{aligned} \therefore I_{CQ} &= \beta \times I_{BQ} \\ &= 8.3 \times 10^{-3} \text{ A} \quad \underline{\underline{\text{Ans.}}} \end{aligned}$$

Applying KVL in O/P,

$$-9 + 7.9 \times 10^3 \times I_{CQ} + V_{CEQ} = 0$$

$$\Rightarrow V_{CEQ} = -56.57 \text{ V} \quad \underline{\underline{\text{Ans.}}}$$

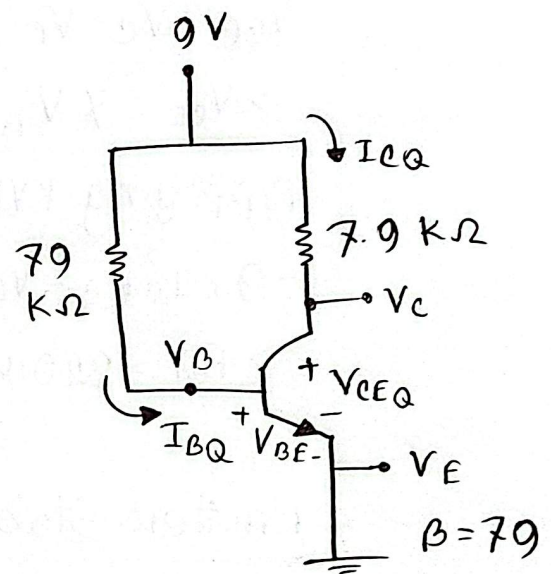
$$V_E = 0 \quad \underline{\underline{\text{Ans.}}}$$

$$\therefore V_C = V_{CEQ} + V_E = -56.57 \text{ V} \quad \underline{\underline{\text{Ans.}}}$$

Again,

$$V_{BE} = V_B - V_E$$

$$\Rightarrow V_B = 0.7 \text{ V} \quad \underline{\underline{\text{Ans.}}}$$



Problem-02

Applying KVL in I/P,

$$-9 + R_B \times I_B + V_{BE} = 0$$

$$\Rightarrow R_B = 105.06 \times 10^3 \Omega \quad \underline{\text{Ans.}}$$

$$I_C = \beta \times I_B = 6.241 \times 10^{-3} \text{ A} \quad \underline{\text{Ans.}}$$

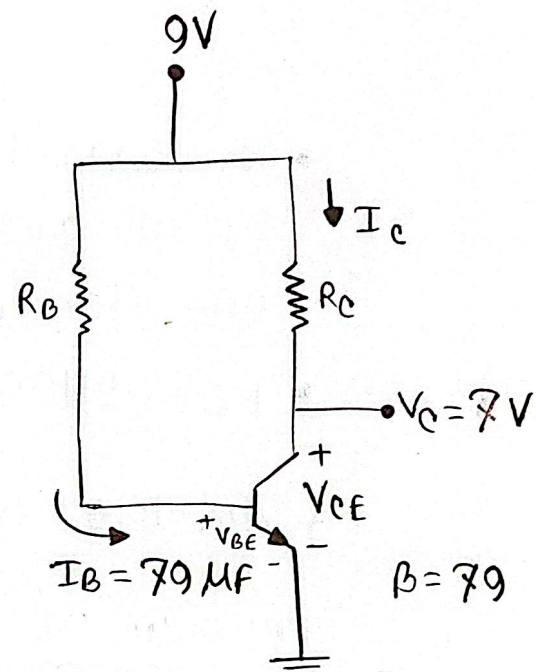
$$V_{CE} = V_C - V_E$$

$$\Rightarrow V_{CE} = 7 \text{ V} \quad \underline{\text{Ans.}}$$

Applying KVL in O/P,

$$-9 + I_C R_C + V_{CE} = 0$$

$$\Rightarrow R_C = 320.46 \Omega \quad \underline{\text{Ans.}}$$



Emitter-Stabilized Bias (4.4)

Problem-01

Applying KVL in I/P,

$$-9 + 79 \times 10^3 \times I_{BQ} + V_{BE} + 7 \times 10^3 \times I_{EQ} = 0$$

$$\Rightarrow -9 + 79 \times 10^3 \times I_{BQ} + V_{BE} + 7 \times 10^3 \times (\beta + 1) I_{BQ} = 0$$

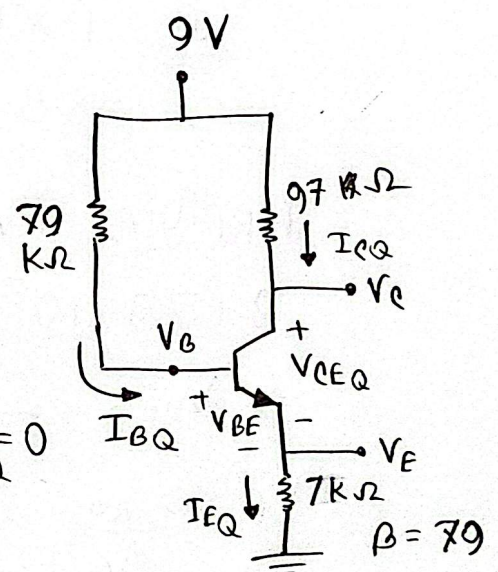
$$\Rightarrow I_{BQ} = 1.29 \times 10^{-5} \text{ A} \quad \underline{\text{Ans.}}$$

$$\therefore I_{CQ} = \beta \times I_{BQ} = 1.02 \times 10^{-3} \text{ A} \quad \underline{\text{Ans.}}$$

Applying KVL in O/P,

$$-9 + 97 \times 10^3 \times I_{CQ} + V_{CEQ} + I_E \times 7 \times 10^3 = 0$$

$$\Rightarrow V_{CEQ} = 1.677 \text{ V} \quad \underline{\text{Ans.}}$$



Now,

$$V_C = 9 - 97 \times I_{CQ}$$
$$= 8.9 \text{ V} \quad \underline{\text{Ans.}}$$

$$V_E = V_C - V_{CEQ}$$
$$= 7.223 \text{ V} \quad \underline{\text{Ans.}}$$

$$V_B = V_{BE} + V_E$$
$$= 7.923 \text{ V} \quad \underline{\text{Ans.}}$$

Problem-02

$$I_B = I_C / \beta = 7.22 \times 10^{-5} \text{ A}$$

Applying KVL in I/P,

$$-9 + I_B \times R_B + V_{BE} + 7 = 0$$

$$\Rightarrow R_B = 18.01 \times 10^3 \Omega \quad \underline{\text{Ans.}}$$

$$7 = I_E \times R_E$$

$$\Rightarrow 7 = (\beta + 1) I_B \times R_E$$

$$\Rightarrow R_E = 989.31 \Omega \quad \underline{\text{Ans.}}$$

$$7.9 = 9 - R_C \times 7 \times 10^{-3}$$

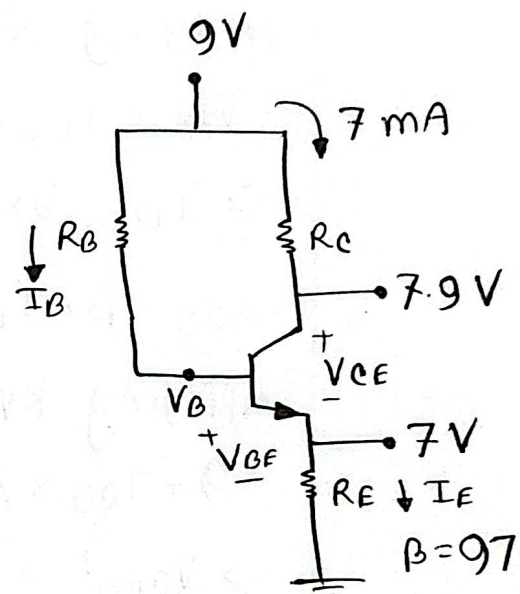
$$\Rightarrow R_C = 157.14 \Omega \quad \underline{\text{Ans.}}$$

$$V_{BE} = V_B - V_E$$

$$\Rightarrow V_B = V_{BE} + V_E = 7.7 \text{ V} \quad \underline{\text{Ans.}}$$

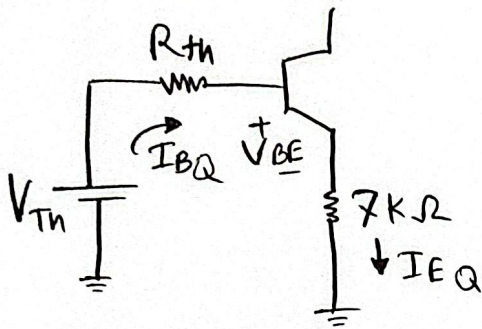
$$V_{CE} = V_C - V_E$$

$$= 0.9 \text{ V} \quad \underline{\text{Ans.}}$$

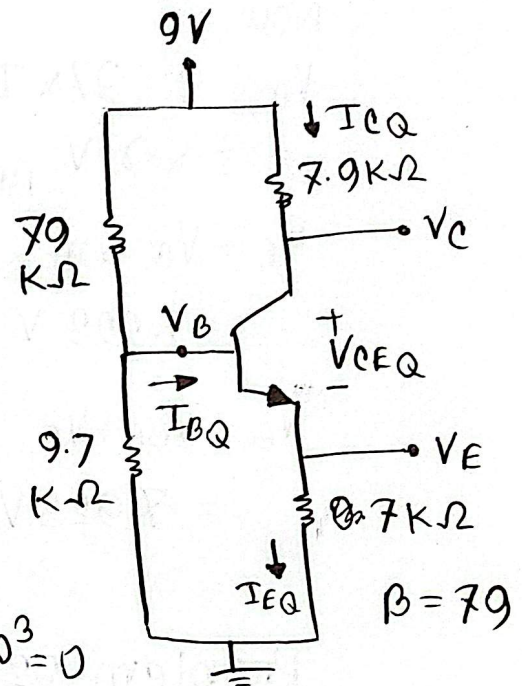


Voltage-Divider Bias (4.5)

Problem - 01



Here,
 $V_{Th} = 0.98 \text{ V}$
 $R_{Th} = 8.64 \text{ k}\Omega$



Applying KVL in I/P,

$$-V_{Th} + I_{BQ} \times R_{Th} + V_{BE} + (\beta + 1)I_{BQ} \times 7 \times 10^3 = 0$$

$$\Rightarrow I_{BQ} = 5 \times 10^{-7} \text{ A} \quad \underline{\underline{\text{Ans.}}}$$

$$I_{CQ} = \beta \times I_{BQ} = 3.95 \times 10^{-5} \text{ A} \quad \underline{\underline{\text{Ans.}}}$$

Applying KVL in O/P,

$$-9 + I_{CQ} \times 7.9 \times 10^3 + V_{CEQ} + (\beta + 1)I_{BQ} \times 7 \times 10^3 = 0$$

$$\Rightarrow V_{CEQ} = 8.41 \text{ V} \quad \underline{\underline{\text{Ans.}}}$$

$$\begin{aligned} V_E &= I_{EQ} \times 7 \times 10^3 \\ &= 0.28 \text{ V} \quad \underline{\underline{\text{Ans.}}} \end{aligned}$$

$$\begin{aligned} V_C &= V_{CEQ} + V_E \\ &= 8.69 \text{ V} \quad \underline{\underline{\text{Ans.}}} \end{aligned}$$

$$\begin{aligned} V_B &= V_{BE} + V_E \\ &= 0.98 \text{ V} \quad \underline{\underline{\text{Ans.}}} \end{aligned}$$

Problem-02

$$V_C = 9 - I_C \times 9.7 \times 10^3$$

$$\Rightarrow I_C = 2.06 \times 10^{-4} \text{ A} \quad \underline{\text{Ans.}}$$

Now,

$$\begin{aligned} V_E &= I_E \times R_E \approx I_C \times R_E \\ &= 2.06 \times 10^{-4} \times 7 \times 10^3 \\ &= 1.44 \text{ V} \quad \underline{\text{Ans.}} \end{aligned}$$

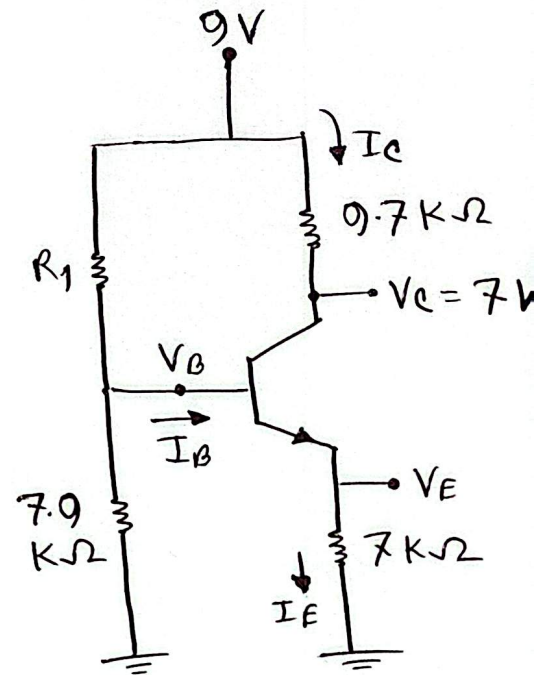
$$V_{BE} = V_B - V_E$$

$$\Rightarrow V_B = V_{BE} + V_E = 2.14 \text{ V} \quad \underline{\text{Ans.}}$$

Again,

$$V_B = \frac{7.9}{7.9 + R_1} \times 9$$

$$\Rightarrow R_1 = 25.32 \text{ K}\Omega \quad \underline{\text{Ans.}}$$



Collector-Feedback Bias (4.6)

Problem-01

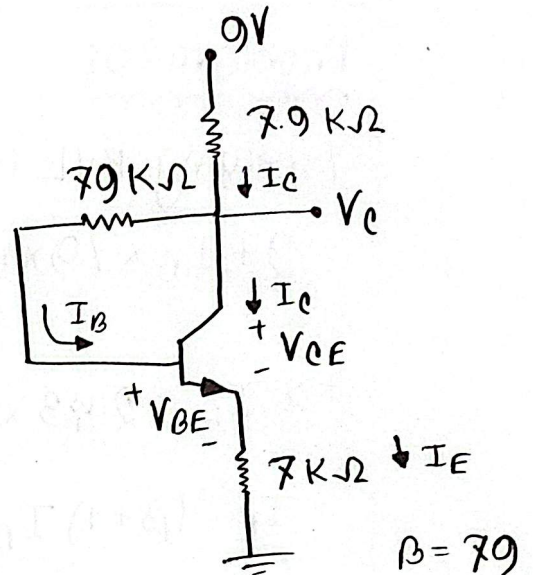
Applying KVL in I/P,

$$-9 + I_B \times 79 \times 10^3 + V_{BE} + (\beta + 1) I_B \times 7 \times 10^3 = 0$$

$$\Rightarrow I_B = 6.53 \times 10^{-6} \text{ A} \quad \underline{\text{Ans.}}$$

$$\therefore I_C = \beta \times I_B = 5.158 \times 10^{-4} \text{ A} \quad \underline{\text{Ans.}}$$

$$\begin{aligned} \therefore V_C &= 9 - I_C \times 79 \times 10^3 \\ &= 4.92 \text{ V} \quad \underline{\text{Ans.}} \end{aligned}$$



Problem-02

Applying KVL in I/P,

$$-9 + I_C \times 9.7 \times 10^3 + (79 + 97) \times 10^3 \times \frac{I_C}{\beta} + V_{BE} + 7.9 \times 10^3 \times (\beta + 1) \frac{I_C}{\beta} = 0$$

$$\Rightarrow I_C = 4.17 \times 10^{-4} \text{ A} \quad \underline{\text{Ans.}}$$

$$\begin{aligned} \therefore V_C &= 9 - I_C \times 9.7 \times 10^3 \\ &= 4.955 \text{ V} \quad \underline{\text{Ans.}} \end{aligned}$$

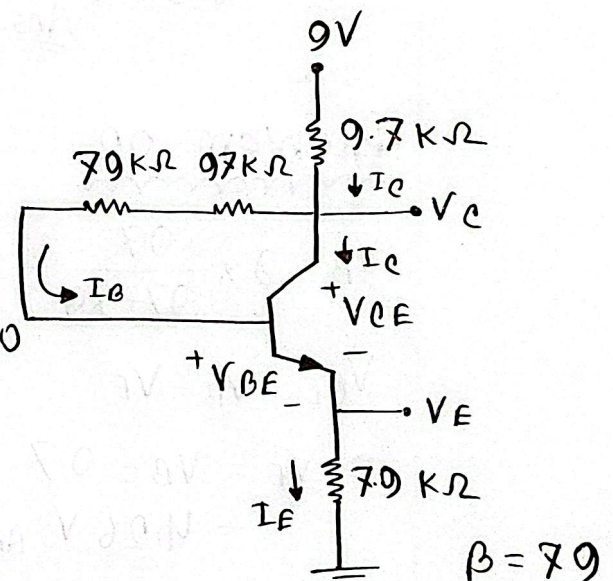
Applying KVL in O/P,

$$-9 + 9.7 \times 10^3 \times I_C + V_{CE} + I_E \times 7.9 \times 10^3 = 0$$

$$\Rightarrow V_{CE} = 1.6191 \text{ V} \quad \underline{\text{Ans.}}$$

$$V_{CE} = V_C - V_E$$

$$\begin{aligned} \Rightarrow V_E &= V_C - V_{CE} \\ &= 3.3359 \text{ V} \quad \underline{\text{Ans.}} \end{aligned}$$



Emitter-Follower Bias (4.7)

Problem-01

Applying KVL in I/P,

$$-9 + I_B \times 79 \times 10^3 + V_{BE} + (\beta + 1) I_B \times 7.9 \times 10^3 - 9 = 0$$

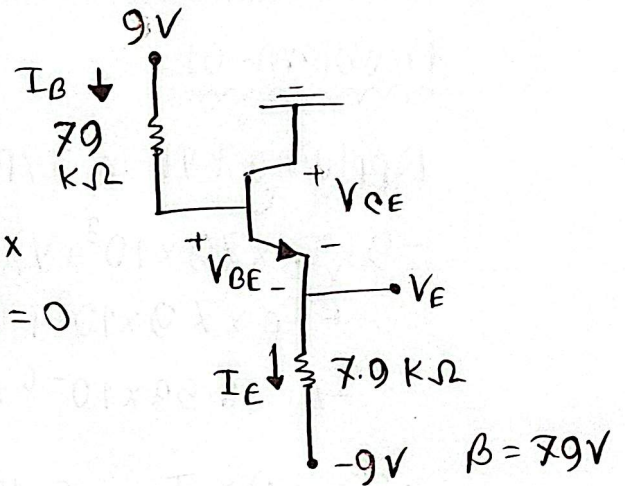
$$\Rightarrow I_B = 2.43 \times 10^{-5} \text{ A}$$

$$\therefore I_E = (\beta + 1) I_B$$

$$= 1.95 \times 10^{-3} \text{ A} \quad \text{Ans.}$$

Applying KVL in O/P,

$$\therefore V_E = I_E \times 7.9 \times 10^3 - 9$$
$$= 6.38 \text{ V} \quad \text{Ans.}$$



Problem-02

$$V_B = 9 \times \frac{97}{97 + 79} = 4.96 \text{ V} \quad \text{Ans.}$$

$$V_{BE} = V_B - V_E$$

$$\Rightarrow V_E = V_B - 0.7$$
$$= 4.26 \text{ V} \quad \text{Ans.}$$

$$V_C = 9 \text{ V} \quad \text{Ans.}$$

Now,

$$V_E = I_E \times 7.9 \times 10^3$$

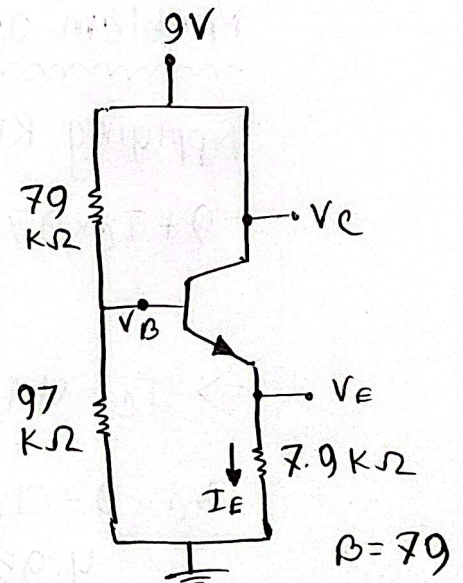
$$\Rightarrow I_E = 5.39 \times 10^{-4} \text{ A} \quad \text{Ans.}$$

$$\therefore I_B = (\beta + 1) I_E = 0.043 \text{ A}$$

$$\therefore I_C = \beta I_B = 3.408 \text{ A}$$

$$\therefore I_E = (\beta + 1) I_B$$

$$\Rightarrow I_B = 6.74 \times 10^{-6} \text{ A} \quad \text{Ans.}$$



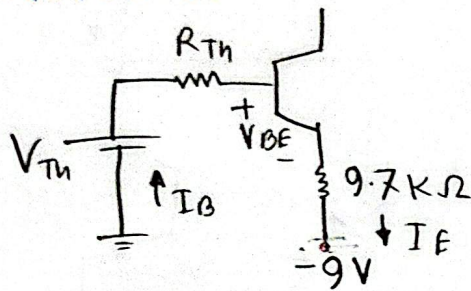
$$I_C = \beta I_B$$
$$= 5.325 \times 10^{-4} \text{ A} \quad \text{Ans.}$$

$$V_{BC} = V_B - V_C = -4.04 \text{ V} \quad \text{Ans.}$$

$$V_{CE} = V_C - V_E = 4.74 \text{ V} \quad \text{Ans.}$$

Miscellaneous Bias configurations (4.9)

Problem-01



Here,
 $V_{Th} = 0V$
 $R_{Th} = 39.9 k\Omega$

Applying KVL in I/P,

$$-V_{Th} + I_B \times R_{Th} + V_{BE} + (\beta + 1) I_B \times 9.7 \times 10^3 - 9 = 0$$

$$\Rightarrow I_B = 1.018 \times 10^{-5} A \quad \underline{\text{Ans.}}$$

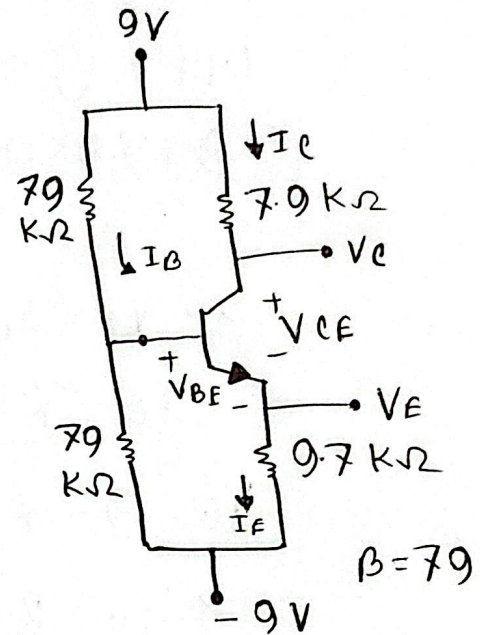
$$\therefore I_C = \beta \times I_B = 8.04 \times 10^{-4} A \quad \underline{\text{Ans.}}$$

$$\therefore I_E = (\beta + 1) I_B = 8.14 \times 10^{-4} A$$

$$\begin{aligned} \therefore V_E &= I_E \times 9.7 \times 10^3 - 9 \\ &= \cancel{7.8958 V} - 1.1 V \quad \underline{\text{Ans.}} \end{aligned}$$

$$\begin{aligned} \therefore V_C &= 9 - I_C \times 7.9 \times 10^3 \\ &= 2.6484 V \end{aligned}$$

$$\begin{aligned} \therefore V_{CE} &= V_C - V_E \\ &= \cancel{5.2474 V} \\ &= 3.75 V \quad \underline{\text{Ans.}} \end{aligned}$$



Problem-02

Applying KVL in I/P,

$$-9 + 7.9 \times 10^3 \times I_C + I_B \times 79 \times 10^3 + V_{BE} = 0$$

$$\Rightarrow -V_C + I_B \times 79 \times 10^3 + V_{BE} = 0$$

$$\Rightarrow I_B = 7.97 \times 10^{-5} \text{ A} \quad \underline{\text{Ans.}}$$

$$V_C = 9 - 7.9 \times 10^3 \times I_C$$

$$\Rightarrow I_C = 2.53 \times 10^{-4} \text{ A} \quad \underline{\text{Ans.}}$$

$$\therefore \beta = \frac{I_C}{I_B} = 3.176 \quad \underline{\text{Ans.}}$$

$$\therefore V_{CE} = V_C - V_E = V_C = 7 \text{ V} \quad \underline{\text{Ans.}}$$

